Statistical Inference

> *“Statistics is the grammar of science.”* – Karl Pearson

Inference vs. Prediction

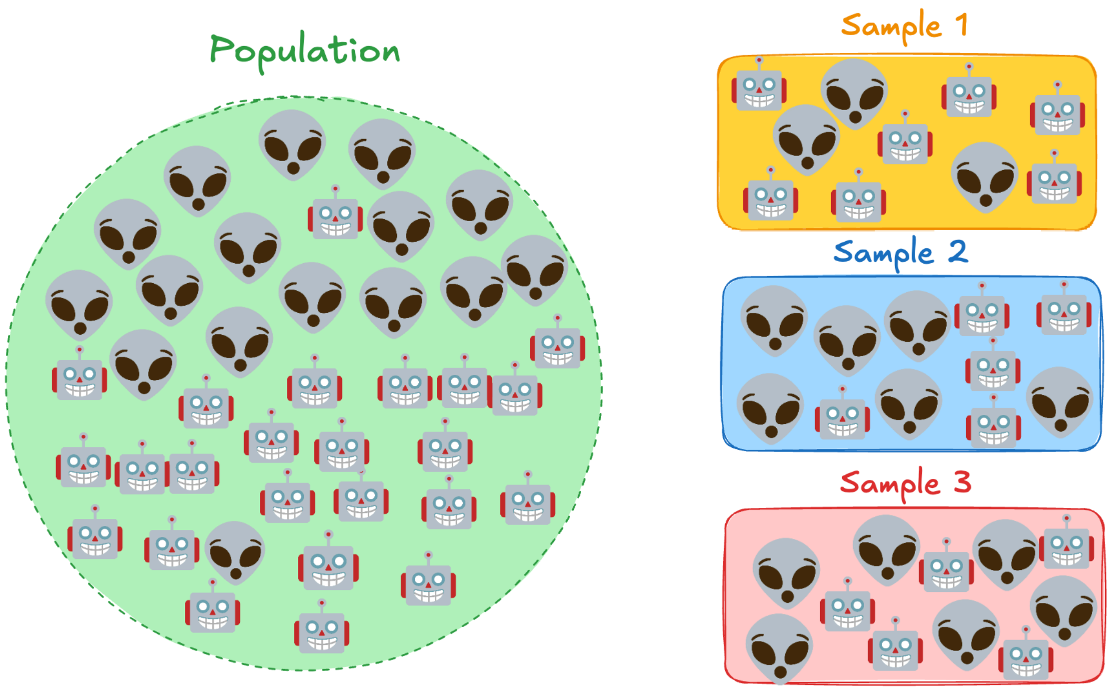
- **Prediction:** use the data to make accurate guesses for new cases
- **Inference:** use the data to support a conclusion about an unknown pattern, parameter, or relationship
- A good prediction can work even when we do not fully understand why it works
- A good inference must explain what claim is being tested, what uncertainty remains, and how strong the evidence is
- The key question changes:
    - Prediction: *How accurate is the answer on new data?*
    - Inference: *What conclusion are we allowed to draw from this data?*

Bayesian vs. Frequentist

- Bayesian: parameters are random but data are fixed
  Uncertainty directly available from the posterior distribution
- Frequentist: true parameters fixed but unknown
  Uncertainty as a result from having finite samples

Population and Sample

What is the proportion of happy robots in the population, **if you only have access to one of the samples**?



Note: If you are able to acquire three sample sequentially, you can use the mark and recapture technique to estimate the size of the population.

Sampling distributions

The distribution of a statistic under repeated sampling from the same population using the same sampling mechanism

- Acknowledge uncertainty of evidence from data
- Foundation of statistical reasoning
- Key to proper statistical inference

Example: Central Limit Theorem and Sampling Distribution

Statistical Reasoning

Given that the observations are finite samples from a true distribution...

- Confidence Interval
  A set of estimators that guarantee chosen coverage probability.
- Hypothesis Testing
  Assuming the null hypothesis, is the dataset an extreme case?
- Multiple Testing
  If one tests infinite number of hypotheses, one of them might be positive by chance.
  xkcd#882 and explainxkcd

Type I Error and the Magical Number 0.05

- A Type I error is a false significant result: we reject the null hypothesis even though the null is true

- If the null is true and we use $p < 0.05$, each independent test has a 5% false positive rate

- With 10 independent null tests, the expected number of false significant results is $10 \times 0.05 = 0.5$

- With 20 independent null tests, the expected number of false significant results is $20 \times 0.05 = 1$

- This is why false discoveries need to be controlled: they can waste time, money, and scientific resources

So what do we think about 5% in the era of big data and big models?

The Magical Number 0.05: Additional References

- The Magical Number Seven, Plus or Minus Two
- BASP banned NHSTP/p-values
- ASA statement on p-values
- Analysis of BASP after p-value ban
- The value of p
- Political Analysis banned p-value

Other Common Tests

- Model-based Tests

- Rank-based Tests

- Simulation-based Tests

Rank-based Tests

A two-sample rank test pools both groups, sorts all observations, and compares ranks instead of raw values.

Use the companion-note widget to change the outlier and watch how raw values become ranks.

Bottom-line: ranks do not ignore outliers. They keep the outlier largest, but stop its raw size from taking over the test.

Model-based Tests: Comparing Two Stories

Many model-based tests start with the same classroom question:

- **Reduced model:** the simpler story, without the extra explanation.
- **Full model:** the richer story, with the extra explanation included.
- **Question:** did the full model improve enough that we should believe the extra explanation matters?

Examples of the same idea:

- **F-test / ANOVA:** compare how much prediction error decreases.
- **Likelihood ratio test:** compare how much better the model explains the observed data.
- **Wald test:** compare an estimated effect to its uncertainty.

The technical tests differ, but the intuition is the same: compare two stories, then ask whether the difference is larger than we would expect by chance.

Full vs Reduced Model: The Picture

Suppose we want to know whether a new variable helps.

- Reduced model: predict using the old variables only.
- Full model: predict using the old variables **plus** the new variable.
- Fit both models to the same data.
- Compare their mistakes or their fit.

If the new variable is useless, the full model may still look a little better just by luck.

So the test asks:

> Is the improvement large compared with the random improvement we would expect under the null hypothesis?

That random-improvement pattern is the **null distribution** of the test statistic.

Same Logic, Different Metrics

Different tests use different ways to measure the gap between the full and reduced models.

- **F-test / ANOVA:** uses the drop in residual error.
- **Likelihood ratio test:** uses the gain in log-likelihood.
- **Wald test:** uses the size of one estimated effect relative to its standard error.

For high-school intuition, focus on the shared logic:

1. Start with a null story: the extra variable does not help.
2. Compute a number that measures how different the full model is from the reduced model.
3. Ask how surprising that number would be if the null story were true.

The formula changes from test to test. The habit of reasoning is the same.

Simulation-based Tests

Hypothesis testing hinges on the null distribution of a chosen test statistic.

When the closed-form null distribution is unavailable, we can approximate it.

- Simulation
  - Using Monte-Carlo methods to obtain the null distribution
  - Need to know the *true distribution*
  - Uncommon in practice but require no calculus
- Permutation
  - Shuffling data to break possible relationships
  - Work for specific null hypotheses
  - Limited applications but assumption-less
- Bootstrap
  - Approximate sampling distribution via resampling
  - Require large sample size
  - Computationally expensive but universal

P-Hacking: Hiding the Searches That Failed

- Example: xkcd 882 — Significant

- If someone reports all 100 tests and only 5 are significant, an informed reader can recognize the Type I error problem

- The danger of p-hacking is hiding the 95 non-significant results

- Then the 5 significant results look like real discoveries instead of what we would expect from searching many times

- The real problem is not just searching; it is searching and then reporting only the exciting results

Optional Study Report: Multiple Testing

- Build a short HTML report showing how false positives accumulate across many tests.
- Specify the null p-value model, number of tests, number of repetitions, reporting rules, and correction rules.
- Explain the difference between honest reporting and hidden-search p-hacking.
- The report must be readable enough that someone can reproduce the simulation from the description and code.